

Experiment 1: Advanced 8-bit ALU with NZCV flags

1. Objective

To design, simulate, and implement an Advanced 8-bit Arithmetic Logic Unit (ALU) capable of performing various arithmetic and logical operations while generating Negative (N), Zero (Z), Carry (C), and Overflow (V) flags.

2. Block Diagram

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``mermaid flowchart TD
  A[Operand A 8-bit] --> ALU[8-bit ALU]
  B[Operand B 8-bit] --> ALU
  Op[Opcode 4-bit] --> ALU
  ALU --> Res[Result 8-bit]
  ALU --> Flags[NZCV Flags 4-bit]
  
```

```

subgraph ALU_Block
  ALU
end
  
```

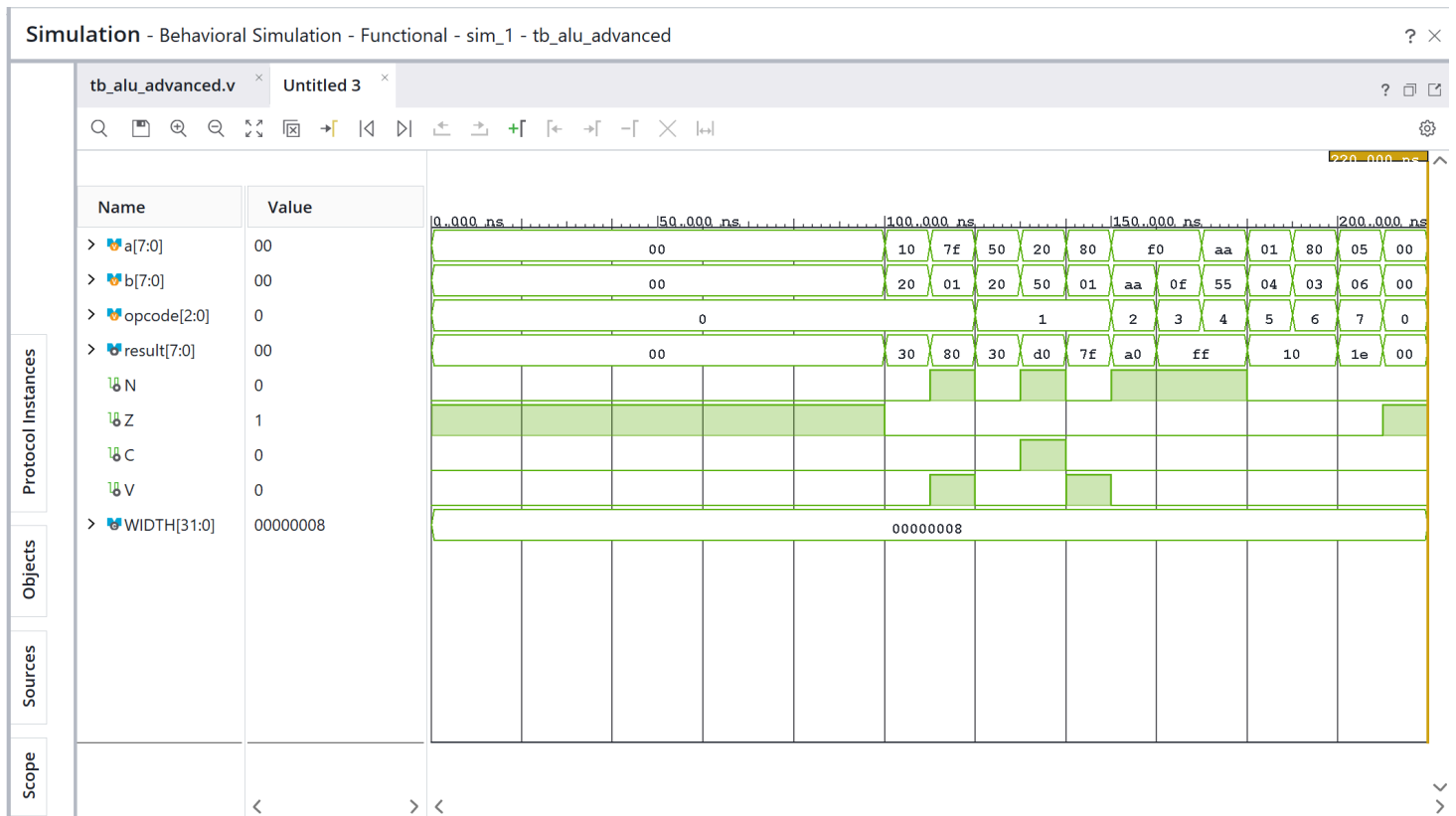
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3. RTL Design

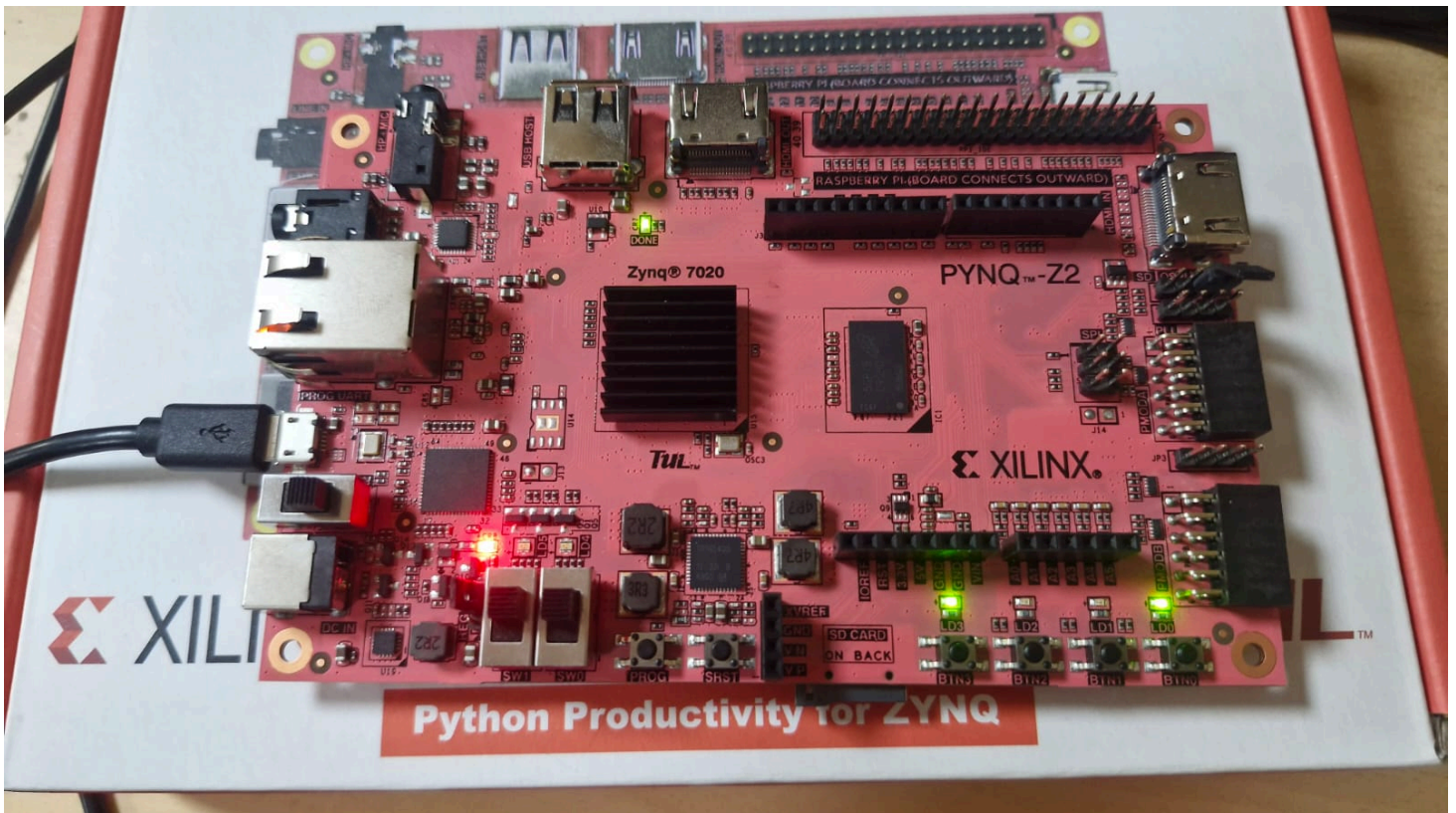
The design consists of an 8-bit ALU that supports operations such as ADD, SUB, AND, OR, XOR, NOT, and shifts. - **8-bit ALU**: The core logic performs the selected operation based on a 4-bit Opcode. - **NZCV Flags**: - **N (Negative)**: Set if the MSB of the result is 1. - **Z (Zero)**: Set if the result is exactly zero. - **C (Carry)**: Set if an arithmetic operation generates a carry out from the MSB. - **V (Overflow)**: Set if an arithmetic operation results in a signed overflow. - **PYNQ-Z2 Auto-Test Wrapper**: A wrapper module designed for the PYNQ-Z2 board to facilitate automated testing by mapping the ALU I/O to the board's switches, buttons, and LEDs for hardware validation.

4. Simulation Results



- Screenshot 1: Addition and Subtraction with N and C flags. - Screenshot 2: Logical operations with N and Z flags. - Screenshot 3: Overflow condition during signed arithmetic.

5. Hardware Implementation



- Photo 1: PYNQ-Z2 board showing ALU inputs via switches. - Photo 2: PYNQ-Z2 board showing output result and flags on LEDs.

6. Applications

- Central Processing Unit (CPU) ALUs
- Digital Signal Processing (DSP) applications
- Embedded microcontrollers

7. Conclusion

An 8-bit Advanced ALU with accurate NZCV flag generation was successfully designed and tested. The implementation on the PYNQ-Z2 board verifies the hardware functionality, making it a reliable building block for more complex computational architectures.